

Feedback on Feedback

Do you know how much energy you consume in your home on a daily basis? Do you know what your appliances use relative to each other, so that you can choose to use the energy-hogging appliances sparingly? You might, but for most consumers, the answer to both of these questions is no. Most residents only receive feedback on their energy use in the form of a monthly bill from their utility provider, making it difficult to identify the biggest sources of energy use in their homes or to take effective action to lower future energy bills.

Too Much Information

Since the 1970s, many researchers from various fields have studied how feedback on energy use affects residential consumer understanding and behavior. Studies involving informative billing and periodic written feedback have realized energy savings between 10% and 20%. If residential consumers had more detailed or more frequent information about their consumption, these studies suggest they would both better understand their energy use patterns and be able to change them effectively. Taken to the extreme, this assumption predicts that the maximum amount of information, delivered continuously, will help residents to maximize their energy savings.

The handful of studies examining continuous energy use feedback in a residential setting seems to support the opposite argument—more information is not always better. First, it seems that real-time feedback does not stimulate more energy conservation compared to monthly or weekly feedback. Indeed, it seems that the major impact of maximizing feedback frequency is increased awareness, not behavioral change or financial savings. Second, an increase in sophistication of real-time feedback technology over the years has not corresponded with an increase in measured



ENERGY, INC.

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energy savings. In fact, one recent study showed less energy savings than were shown in a study using a basic electricity monitor 30 years ago!

Making It Real Time

As a college student, I conducted a ten-household pilot study to explore this issue further. I wondered how modern households would react to a simple, nongraphic real-time display of electricity use, similar in appearance to those developed in the '70s. Would this simpler real-time feedback be effective in helping residents understand and change their energy use habits?

The monitor I chose for this study was The Energy Detective (TED), a device that measures whole-house electricity use. It consists of a 6-inch countertop display unit that can be plugged into any house outlet (see photo). Once the monitor has been plugged in and

electricity rates have been programmed, the monitor immediately displays current kW use and dollars spent per hour on electricity. When TED is working correctly, a green LED flashes once per second. Using two buttons, users can display six different attributes of their electricity use in either dollars or kWh, and access several other features, including the monthly bill, number of days left in a billing cycle, voltage in real time, and a standard timer. By pressing buttons in combination, users can program a visual or auditory alarm to go off when a usage goal has been exceeded. Users can also edit electricity rate information or check their historical use.

I chose The Energy Detective for this study because it is a relatively cheap (\$140), commercially available monitor with marketplace success. There are no graphic displays or colorful, flashy features, except for the intermittently blinking green LED. In essence, this

monitor has better information features than monitors of the past, without the added graphic and manipulative capabilities available in computer energy monitors.

I installed The Energy Detectives in ten households—five in a low-income neighborhood and five in a high-income neighborhood—in Oberlin, Ohio, during January 2006. After I installed each monitor, I programmed it in the presence of the homeowner, and then demonstrated how to switch between different screens. Homeowners were welcome to ask questions, and were left with a product manual and savings chart. Homeowners were given no specific advice on using the monitors to save energy. They were simply told to use the monitors as they liked, and that we would return a month later to talk with them about their experience.

In order to assess whether the ten households were saving electricity, I obtained utility bill data, with permission, from these households and from 50 additional households in the two neighborhoods. These 50 neighborhood households had participated in a door-to-door energy use survey, but were not contacted further or told to try to conserve energy; these households formed the control group for the study.

Between January and March 2006, per capita electricity use reduction did not differ significantly between the ten-household sample and the control group. However, the largest baseline user in the sample did show significant energy savings. This household was motivated by the information they got from TED to turn more things off at night, for instance.

Monthly interviews revealed that the majority of sample households did not use the monitor as frequently or as intensely as they could have done and/or had planned to, with the exception of two families. The degree to which households used the monitor was somewhat correlated with relative baseline per capita electricity use; low-baseline users tended to use the monitor less often, and the only household with

energy savings approaching significance was a very high-baseline user. With one exception, the low-income households used the monitor far less than the higher-income households.

All but two households reported feeling an effect from the simple presence of the monitor. Additionally, most households looked at the home screen on a regular basis, and reported being more aware of the energy use of different appliances.

Ease of Use

Few households investigated different screens or special features, such as the alarm mode, history, timer, or even month-to-date usage. It seems that while most households agreed that the “home screen” was easy to understand, it would have been necessary to take the time to consult the TED manual in order to use the other screens and features.

When homeowners did explore different screens, there was considerable confusion as to what the numbers actually meant. None of the households attempted to use the special alarm or history features of TED. This was probably because these features can be used only by pressing buttons in combination, and it is nearly impossible to figure out which buttons to press unless one reads the manual.

Homeowners had clear ideas as to how the monitor could be improved. Two households would have preferred graphic displays that synthesized the numerical information on the screen in a way that allowed them to analyze their energy use over time with a single glance. Five households felt that the monitor would have been more helpful if they had been able to limit the functions available to them. Homeowners agreed that there were too many screen options, and the high ratio of monitor functions to buttons made navigation confusing.

Although significant energy savings were not realized, five out of ten households opted to keep the monitor.

The monitor became an interesting source of information for these households, and they did not want to lose the increased awareness of energy use that they had gained from keeping the monitor in their homes. So a simple monitor such as TED is clearly valued as an educational tool, but it does not seem likely to pay for its costs in energy savings. Committed conservers might be able to use the monitor to reduce energy use over time, but only one of our households fell into this group.

Only one out of ten households felt that the monitor would save them enough money to pay for itself over time. Even more surprising is the fact that half the households chose not to keep the monitor when it was offered to them for free, indicating that these households did not even value it as an educational tool. In other words, there is a lot of work to be done before this type of product is marketable to the average residential energy consumer.



—Daisy Allen

Daisy Allen is a recent graduate of Oberlin College, in Oberlin, Ohio, where she received her BA in Environmental Studies and Economics. She is now a research intern for KEMA, Incorporated.

For more information:

For more information about this project, contact the author by e-mail: daisy.allen@kema.com.

Energy, Incorporated, the company that manufactures The Energy Detective, has released a new version of its product, the Energy Detective 5000. In this version, the monitor's data are downloadable, and customers can view historical data in a graphic format. For more information, go to www.theenergydetective.com.